

Name: _____

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1. Let $a_n = (-1)^n$. Prove that $\{a_n\}_{n=1}^{\infty}$ diverges.
2. For each $n \in \mathbb{N}$, let $a_n = 2023 - \frac{1}{\sqrt{n}}$. Show that $\{a_n\}_{n=1}^{\infty}$ converges to 2023.
3. Let $\{a_n\}_{n=1}^{\infty}$ be a sequence with the property that $a_n \in \mathbb{Z}$ for every $n \in \mathbb{N}$.
 - (a) Is it possible that a_n converges to 09/28?
 - (b) If $a_n \neq a_m$ for all $n \neq m$, prove that $\{a_n\}_{n=1}^{\infty}$ does not converge.
 - (c) Suppose $\{a_n\}_{n=1}^{\infty}$ converge and that $a_1 = 2023$. What can be said about this sequence?
4. If $\{a_n\}_{n=1}^{\infty}$ converges to 2 and $\{b_n\}_{n=1}^{\infty}$ converges to 23, prove (by definition) that $\{1000a_n + b_n\}_{n=1}^{\infty}$ converges to 2023.
5. For each $n \in \mathbb{N}$, let $a_n = \frac{n^2-1}{n^2}$. Prove that $\{a_n\}_{n=1}^{\infty}$ is a Cauchy sequence.